INSTITUTE OF MANAGEMENT SCIENCES PESHAWAR

im | sciences

Course Code & Title:	**Software Requirements Engineering**
Credit Hours:	2+1
Program(s) & Group(s):	BSSE
Academic Calendar Semester:	
Lecture Timing:	
Prerequisites and/or Expectations:	Software Engineering.
Instructor:	Mr. Hamood ur Rehman Durrani
Website (if any):	
Email:	hamoodurrehman@imsciences.edu.pk
Office Location:	Ground Floor, Academic Block
Office Contact Hours:	Monday to Friday 8:30 to 4:30
Course Description:	The course will introduce the basics of requirements engineering. It will discuss the issues, processes, and techniques in requirements engineering. The course will teach different modeling approaches in problem solving. It will introduce advance and potential research topics in requirements engineering.
Course Objectives (COs):	1. To understand the role of RE in system and software engineering. 2. To understand and apply the methods and techniques for Requirements Elicitation, Analysis, Specification, Verification and Validations within the RE Process.

Course Learning Outcomes (CLOs):

At the end of the course the students will be able to:	**Domain**	**BT Level**
1. Describe the requirements engineering process 2. Effectively elicit and analyze software requirements for the development of cost-effective and efficient technical solutions. 3. Validate & document effectively both functional and non-functional requirements in Software Requirements Specification (SRS) using clear, unambiguous requirements.		Explain Analyze & Apply Evaluate

* BT= Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A= Affective domain

Alignment of CLOs and COs

	CO-1	CO-2	CO-3	CO-4	CO-5
CLO-1	X	X			
CLO-2	X	X			
CLO-3	X	X			

<table>
<tr><td>Course Assessment(s):</td><td colspan="3"><u>As per IMSciences policy</u>
• Mid-term Exam (30%)
• Assignment (20%)
• Final-term Exam (50%)

<table><tr><th>S.No.</th><th>CLOs</th><th>Relevant Assessment Methods</th></tr><tr><td>1.</td><td>1, 2, 3, 4</td><td>Mid-term & Final-term Exam</td></tr><tr><td>2.</td><td>1, 2, 3, 4</td><td>Quizzes, assignments & Semester Project</td></tr></table></td></tr>
<tr><td>Graduate Attributes (GAs)/ Program Learning Outcomes (PLOs)</td><td colspan="3">1. Gain an understanding of the underpinning theories of fundamental principles and technologies within the area of computer science (Academic education).
2. Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements (Knowledge for Solving Computing Problems).
3. Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines (Problem Analysis).
4. Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations (Design/ Development of Solutions)
5. Create, select, adapt, and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations (Modern Tool Usage)
6. Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings (Individual and Teamwork)
7. Communicate effectively with the computing community and with society about complex computing activities by being able to comprehend and write effective reports, instructions (Communication)
8. Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice (Computing Professionalism and Society)
9. Understand and commit to professional ethics, responsibilities, and norms of professional computing practice (Ethics)
10. Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional (Life-long Learning)</td></tr>
<tr><td>Alignment of CLOs and PLOs/GAs</td><td colspan="3"><table><tr><th></th><th>GA-1</th><th>GA-2</th><th>GA-3</th><th>GA-4</th><th>GA-5</th><th>GA-6</th><th>GA-7</th><th>GA-8</th><th>GA-9</th><th>GA-10</th></tr><tr><td>CLO-1</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td>X</td><td></td><td></td><td></td></tr><tr><td>CLO-2</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td>X</td><td></td><td></td><td></td></tr></table></td></tr>
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	CLO-3			X	X			X			
Course Resources:	1. Book Title: Requirements Engineering: Processes and Techniques Citation: Kotonya and Sommerville (1998) Author: Gerald Kotonya and Ian Sommerville Edition:1st Edition ISBN: 978-0-471-97208-2 Publisher:Wiley Software Requirements, Wiegers K. &Beatty J., 3rd Ed. Microsoft Press 2. Book Title: Software Requirements: Objects, States, and Functions, Citation: Davis (1993), Author: Alan Davis, Edition: 2nd Edition, ISBN: 978-0138057633, Publisher: Prentice Hall 3. Book Title: Software Engineering: A Practitioner's Approach, Citation: Pressman (2001), Author: Roger Pressman, Edition: 5th Edition, Publisher: McGraw Hill										
Course Teaching Methodology:	This course employs a blended teaching methodology that combines interactive lectures with hands-on practical exercises to reinforce learning. Students will engage with real-world case studies to apply analysis and design principles directly. Frequent use of diagrams in ensures practical understanding, while peer discussions and project-based assessments foster critical thinking and collaborative skills. This approach ensures a balanced acquisition of theoretical knowledge and practical application.										
Course Assignment/Quizzes /Presentation:	• Students are required to submit different milestones of a semester project for technical implantation of the methods and tools taught during the class. • There can be multiple quizzes during the semester which can either be announced in advance or can be surprised.										
Attendance Policy	As per the IMSciences policy students will not be allowed to sit in the final-Term exam if they fail to attend less than 75% of the total classes. Attendance will be marked after 15 minutes of the class starts. If a student comes into the class after 15 minutes, he/she may sit in the class, but their attendance will not be marked for that class. In case of emergencies or unavoidable circumstances students have to report me directly or e-mail.										
General Policies and Behavioral Expectations:	During the class sessions students are expected to. • Ensure timely arrival in class and remain present throughout the class session. • Participate in class discussions and engage in any individual or group tasks. • Adhere to institute attendance policy and general code of conduct. • Maintain and update record of all class notes, handouts, and relevant materials. • Adhere to basic principles of academic integrity with regards to exams and assignments • Use the lab equipment properly and carefully.										

	• Students are not allowed to play games or use any offline/online application or other media (news, social media, movies etc.) in computer labs/classroom. • Students must follow all the cyber laws and maintain other's privacy while working in computer labs

Grades to be Assigned

Grade	Marks (%)	Grade Points	Remarks
A+	91-100	4.0	Outstanding
A	87-90	4.0	Excellent
B+	80-86	3.5	Very Good
B	72-79	3.0	Good
C+	66-71	2.5	Satisfactory
C	60-65	2.0	Pass
F	Below 60	0	Fail

Week wise distribution

Week No	Topic Description	CLOs
1	• Introduction • Understanding Software Requirements • Kinds of Requirements-Functional Requirements	CLO-1, CLO-3
2	• Non-Functional Requirements • Product requirements • Organizational requirements • External requirements	CLO-1, CLO-3
3	• Measurement and Metrics for Non-Functional Requirements • Domain Requirements • Inverse requirements • Design and implementation constraints	CLO-2, CLO-3
4	• User/Customer Requirements • System Contract Requirements • Requirements Problems	CLO-2, CLO-3
5	• Requirements Engineering (RE) Process-Introduction • RE Process-Inputs and Outputs • RE Process Variability Factors • Requirements Engineering Process Activities	CLO-1
6	• Actors and stakeholders in RE • Role-Action Method • RE-Process improvement	CLO-1, CLO-2

7	• Social and Cultural Issues in RE-Introduction • Six Areas of Social Issues • Cultural Issues in RE • Addressing Social and Cultural Issues	CLO-1, CLO-2
8	• Requirements Elicitation • Problems in Requirements Elicitation • Contexts in Requirements Elicitation Process • Ethnomethodology	CLO-2
9	• Requirements Elicitation Stages • Basics of Knowledge Acquisition • Knowledge Structuring Techniques	CLO-2
10	• Requirements Elicitation Techniques Interviews • Scenarios • Observations and social analysis • Requirements reuse and Prototyping	CLO-2
11	• Requirements Analysis-Introduction Requirements Analysis Stages • Requirements Analysis Techniques	CLO-2
12	• Requirements Analysis-Introduction • Requirements Analysis Stages • Requirements Analysis Techniques	CLO-2
13	• Requirements Errors & its types • Addressing Requirements Errors	CLO-2, CLO-3
14	• Requirements defects removal- • Ad-hoc review • Checklist review • Defect-based reading • Perspective-based reading	CLO-3
15	• Requirements Validation stage- Inputs and Outputs • Requirements Review Process & Activities • Review checklists • Prototyping • User manual development	CLO-3
16	• Model validation • Requirements testing	CLO-3